E16 Rate determining step

This is example 5.9 from Rawlings Ekerdt.

The synthesis of methane is carried out over a ruthenium catalyst:

$3H_2$ (g) + CO (g) --> CH_4 (g) + H_2O (g)

One possible mechanism is

1) CO (g) + S ⇋ CO_{ads} QE
2) CO_{ads} + S ⇋ C_{ads} + O_{ads} QE
3) O_{ads} + H_2 (g) ⟶ H_2O (g) + S irreversible
4) H_2 (g) + 2S ⇋ $2H_{ads}$ QE
5) C_{ads} + H_{ads} ⇋ CH_{ads} + S QE
6) CH_{ads} + H_{ads} ⇋ $CH_{2,ads}$ + S QE
7) $CH_{2,ads}$ + H_{ads} ⇋ $CH_{3,ads}$ + S QE
8) $CH_{3,ads}$ + H_{ads} ⟶ CH_4 (g) + 2S irreversible, RDS

Step 8 is the RDS. Write the rate law for this reaction in terms of only reactant and product species, i.e., do not include any surface concentrations of intermediate species.

rate law: $r_{CH_4} = k_{8f}\,\theta_{CH_3}\,\theta_H$

θ_{CH_3}: use rxn 7:

$$0 = k_{7f}\,\theta_{CH_2}\,\theta_H - k_{7r}\,\theta_{CH_3}\,\theta_*$$

$$\theta_{CH_3} = K_{c7}\,\frac{\theta_{CH_2}\,\theta_H}{\theta_*}$$

θ_{CH_2}: use rxn 6:

$$0 = k_{6f}\,\theta_{CH}\,\theta_{H} - k_{6r}\,\theta_{CH_2}\,\theta_{*}$$

$$\theta_{CH_2} = K_{C6}\,\frac{\theta_{CH}\,\theta_{H}}{\theta_{*}}$$

θ_{CH}: use rxn 5:

$$0 = k_{5f}\,\theta_{C}\,\theta_{H} - k_{5r}\,\theta_{CH}\,\theta_{*}$$

$$\theta_{CH} = K_{C5}\,\frac{\theta_{C}\,\theta_{H}}{\theta_{*}}$$

θ_{C}: use rxn 2:

$$0 = k_{2f}\,\theta_{CO}\,\theta_{*} - k_{2r}\,\theta_{C}\,\theta_{O}$$

$$\theta_{C} = K_{C2}\,\frac{\theta_{CO}\,\theta_{*}}{\theta_{O}}$$

θ_{CO}: use rxn 1:

$$0 = k_{1f}\,C_{CO}\,\theta_{*} - k_{1r}\,\theta_{CO}$$

$$\theta_{CO} = K_{C1}\,C_{CO}\,\theta_{*}$$

θ_O: Assume $r_3 = r_8$ which is reasonable since H_2O (g) & CH_4 (g) are produced in stoichiometric amounts.

$$k_{3f} \theta_O C_{H_2} = k_{8f} \theta_{CH_3} \theta_H$$

$$\theta_O = \frac{k_{8f}}{k_{3f}} \frac{\theta_{CH_3} \theta_H}{C_{H_2}}$$

restate rate law: $r_{CH_4} = k_{8f} \theta_{CH_3} \theta_H$

develop expression for θ_{CH_3}:

$$\theta_{CH_3} = K_{C7} \frac{\theta_{CH_2} \theta_H}{\theta_*}$$

sub in θ_{CH_2}:

$$\theta_{CH3} = K_{C7} K_{C6} \frac{\theta_{CH} \theta_H^2}{\theta_*^2}$$

sub in θ_{CH}:

$$\theta_{CH3} = K_{C7} K_{C6} K_{C5} \frac{\theta_C \theta_H^3}{\theta_*^3}$$

sub in θ_C:

$$\theta_{CH3} = K_{C7} K_{C6} K_{C5} K_{C2} \frac{\theta_{CO} \theta_H^3}{\theta_O \theta_*^2}$$

sub in θ_O:

$$\theta_{CH3} = K_{C7} K_{C6} K_{C5} K_{C2} \frac{K_{3f}}{K_{8f}} \frac{C_{H2} \theta_{CO} \theta_H^2}{\theta_{CH3} \theta_*^2}$$

re arrange:

$$\theta_{CH3}^2 = K_{C7} K_{C6} K_{C5} K_{C2} \frac{K_{3f}}{K_{8f}} \frac{C_{H2} \theta_{CO} \theta_H^2}{\theta_*^2}$$

sub in θ_{CO} & θ_H:

θ_H: get from rxn 4:

$$0 = k_{4f} C_{H_2} \theta_*^2 - k_{4r} \theta_H^2$$

$$\theta_H = \theta_* \sqrt{K_{c4} C_{H_2}}$$

going back to θ_{CH_3}:

$$\theta_{CH_3}^2 = \frac{k_{3f}}{k_{8f}} K_{c7} K_{c6} K_{c5} K_{c2} C_{H_2}^2 K_{c1} C_{co} K_{c4} \theta_*$$

Combine:

$$k' = \left(\frac{k_{3f}}{k_{8f}} K_{c7} K_{c6} K_{c5} K_{c2} K_{c1} K_{c4} \right)^{1/2}$$

simplify:

$$\theta_{CH_3} = k' C_{H_2} C_{co}^{1/2} \theta_*^{1/2}$$

substitute:

$$r_{CH_4} = k_{8f} \theta_{CH_3} \theta_H$$

$$= k_{8f} k' C_{H_2} \theta_H C_{co}^{1/2} \theta_*^{1/2}$$

Sub in θ_H:

$$r_{CH_4} = k_{rf}\, k' K_{C4}^{1/2} C_{H_2}^{3/2} C_{CO}^{1/2} \theta_*^{3/2}$$

Combine:

$$k'' = k_{rf}\, k' K_{C4}^{1/2}$$

$$r_{CH_4} = k'' C_{H_2}^{3/2} C_{CO}^{1/2} \theta_*^{3/2}$$

θ_*: get from site balance

$$\theta_{tot} = \theta_* + \theta_{CO} + \underbrace{\cancel{\theta_O}}_{0} + \underbrace{\cancel{\theta_H}}_{0} + \underbrace{\cancel{\theta_{CH}}}_{0} + \underbrace{\cancel{\theta_{CH_2}}}_{0}$$

$$+ \underbrace{\cancel{\theta_{CH_3}}}_{0} + \underbrace{\cancel{\theta_C}}_{0}$$ assumptions

then: $\theta_{tot} = \theta_* + K_{C1} P_{CO} \theta_*$

$$\theta_* = \frac{\theta_{tot}}{1 + K_{C1} C_{CO}}$$

assume $\theta_{CO} \gg \theta_*$. Then $K_{C1} C_{CO} \gg 1$, so then:

$$\theta_* = \frac{\theta_{tot}}{K_{C1} C_{CO}}$$

Substitute: $r_{CH_4} = k'' C_{H_2}^{3/2} C_{CO}^{1/2} \theta_*^{3/2}$

$$r_{CH_4} = \frac{k'' \theta_{tot}^{3/2} C_{H_2}^{3/2}}{K_{C1}^{3/2} C_{CO}} = \frac{k_{eff} C_{H_2}^{3/2}}{C_{CO}}$$